

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 10-2024: A 46-Year-Old Woman with Hyperglycemia Refractory to Insulin Therapy

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and Stephen O'Rahilly, M.D.

PRESENTATION OF CASE

Dr. Kaitlyn R. Shank (Medicine): A 46-year-old woman was admitted to this hospital because of hyperglycemia that was refractory to insulin therapy.

Five years before the current presentation, the patient received a diagnosis of type 2 diabetes mellitus after nausea, vomiting, polyuria, and polydipsia had developed. At that time, the weight was 113 kg, and the body-mass index (BMI; the weight in kilograms divided by the square of the height in meters) was 45.6. Treatment with metformin was started, along with subcutaneous insulin glargine at a dose of 10 units daily. The patient had adverse effects associated with metformin, so treatment with metformin was discontinued and glipizide was started. During the next 3.5 years, despite progressive weight loss, the patient's glycemic control worsened, which resulted in gradual intensification of her glucose-lowering regimen.

Two years before the current presentation, the patient moved to the Boston area and established care at a clinic affiliated with another hospital. The weight was 63 kg, and the BMI was 25.4. Medications for diabetes included subcutaneous insulin glargine (20 units daily) and glipizide, and the glycated hemoglobin level was 13.2% (reference range, 4.3 to 5.6). Treatment with empagliflozin was started but then stopped because of a urinary tract infection. The dose of subcutaneous insulin glargine was increased to 30 units daily, and treatment with subcutaneous insulin lispro, administered at a dose of 10 units three times a day with meals, was added. Treatment with glipizide was stopped, and treatment with glimepiride and pioglitazone was started.

One year before the current presentation, the weight was 48 kg, and the BMI was 19.4. Medications for diabetes included subcutaneous insulin glargine (34 units daily), subcutaneous insulin lispro (15 units three times a day with meals), glimepiride, and pioglitazone, and the glycated hemoglobin level was 14.0%. Anorexia had developed, along with fatigue. Although the patient reported eating very little, the blood glucose levels remained markedly elevated.

Nine months before the current presentation, the weight was 42 kg, and the BMI was 16.9. Medications for diabetes included subcutaneous insulin glargine (46 units

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N Engl J Med 2024;390:1219-29.

DOI: 10.1056/NEJMcpc2312729

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daily), subcutaneous insulin lispro (20 units three times a day with meals), glimepiride, and pioglitazone, and the glycated hemoglobin level was 11.0%.

Six months before the current presentation, persistent abdominal pain developed, and the patient was admitted to the other hospital. Computed tomography (CT) of the abdomen and pelvis, performed after the administration of intravenous contrast material, reportedly did not show evidence of obstruction, perforation, or abscess. The abdominal pain was attributed to gastroparesis; treatment with gabapentin, metoclopramide, and oxycodone was started. During the hospitalization, the patient was found to have profound hyperglycemia without diabetic ketoacidosis or hyperosmolar hyperglycemic syndrome. The glycated hemoglobin level was 15.5%. Testing was negative for antibodies to glutamic acid decarboxylase, islet antigen 2, and zinc transporter 8. A diagnosis of lipodystrophy was considered. Treatment with human regular U-500 insulin, given in daily doses of 250 units with breakfast and 200 units each with lunch and dinner, was started; treatment with subcutaneous insulin glargine and subcutaneous insulin lispro was stopped. The patient was discharged home. Subsequent to this hospitalization, a gastric-emptying study reportedly showed normal emptying at 1 hour.

Two months before the current presentation, the abdominal pain worsened, and the patient was again admitted to the other hospital. Her hyperglycemia was initially treated with continuous intravenous insulin at doses of up to 40 units per hour. When the blood glucose level improved, treatment with intravenous insulin was stopped. Treatment with human regular U-500 insulin was then resumed at an increased dose (550 units each with breakfast and lunch and 450 units with dinner); treatment with isophane insulin, also known as neutral protamine Hagedorn (NPH) insulin, was started at a dose of 80 units twice a day. The abdominal pain subsided with improved glycemic control, and the patient was discharged home.

Over the course of the 2 months between hospital discharge and the current presentation, the patient had five additional admissions to the other hospital because of abdominal pain and hyperglycemia. During each admission, the hyperglycemia was treated with high-dose intrave-

nous insulin, the abdominal pain abated with improved blood glucose levels, and the patient was discharged home. During one admission, endoscopy was performed and reportedly revealed erosive gastritis and duodenitis due to infection with *Helicobacter pylori*, for which the patient was treated with amoxicillin, clarithromycin, bismuth subsalicylate, and pantoprazole.

Nine days before the current presentation, the abdominal pain worsened, and the patient was transported by ambulance to the emergency department of the other hospital. Treatment with intravenous insulin was initiated, and the patient was admitted to the hospital. On the second hospital day, treatment with intravenous insulin was stopped, and treatment with human regular U-500 insulin was resumed at an increased dose of 650 units each with breakfast and lunch and 600 units with dinner; treatment with NPH insulin was also restarted at a dose of 80 units twice a day. She was transferred to this hospital for further care.

On arrival at this hospital, the patient reported constant epigastric abdominal pain. She also reported that she had been administering her subcutaneous insulin medications at home without missed doses and confirmed that she had been rotating the injection sites on her abdomen. Other medical history included depression, as well as anemia and leukopenia; a bone marrow biopsy performed 7 months earlier revealed mild-to-moderate fibrosis with patchy myxoid degeneration. She had numbness and tingling in the hands and feet that had begun 9 months before the current presentation. Her menses had stopped 14 months before the current presentation. Other medications included acetaminophen, fluoxetine, gabapentin, glipizide, hydromorphone, lorazepam, mirtazapine, oxycodone, pancrelipase, pantoprazole, and pioglitazone. She had no known drug allergies. The patient was Black and lived in Boston with her husband and children; she had previously worked as a chef but was not working at the time of the current presentation. She rarely drank alcohol or smoked marijuana and had never smoked tobacco; she did not use illicit drugs. Her mother had rheumatoid arthritis and type 2 diabetes, and her father had liver disease.

On examination, the temporal temperature was 36.1°C, the blood pressure 108/78 mm Hg, the

pulse 98 beats per minute, the respiratory rate 16 breaths per minute, and the oxygen saturation 100% while the patient was breathing ambient air. The weight was 42 kg, and the BMI was 17.0. The patient appeared cachectic with a diffuse absence of body fat and muscle mass. Her face had acromegaly-like features, including broadening of the nose and thickening of the lips (Fig. 1). Acanthosis nigricans was present, most notably in the neck and axillae. She had skin tags on the face and neck, and vitiligo was present on the ears and on the fingers of the right hand. The abdomen was soft, nondistended, and diffusely tender on palpation, without rebound or guarding; there was no hepatosplenomegaly. There was skin breakdown in the legs. The remainder of the examination was normal.

Laboratory testing showed a fasting blood glucose level of 311 mg per deciliter (17.3 mmol per liter; reference range, 70 to 100 mg per deciliter [3.9 to 5.6 mmol per liter]) and a C-peptide level of 12.1 ng per milliliter (reference range, 1.1 to 4.4). Blood levels of cholesterol, triglycerides, aspartate aminotransferase, alanine aminotransferase, alkaline phosphatase, total bilirubin, total complement, C3, and C4 were normal. The blood level of leptin was undetectable, and the blood level of adiponectin was 24.5 μ g per

milliliter (reference range, 2.9 to 30.4). A screening test for human immunodeficiency virus was negative. Testing for antibodies to nuclear antigens was positive at a titer of 1:320 with a homogeneous pattern; testing for antibodies to double-stranded DNA was negative. Serum protein electrophoresis showed a normal pattern with a mild diffuse increase in the gamma globulin level. Other laboratory test results are shown in Table 1. Imaging studies were performed.

Dr. Leo L. Tsai: CT of the abdomen and pelvis, performed without the administration of intravenous contrast material, revealed a near-complete absence of both visceral and subcutaneous fat, with multiple skin folds reflecting a clinically significant lack of fat (Fig. 2). There was no bowel dilatation. The pancreas was atrophic, a finding commonly seen in patients with chronic insulin-dependent diabetes.¹ There were bilateral, non-obstructing renal stones measuring up to 3 mm in diameter. The attenuation level of the liver was 58 Hounsfield units (HU), which is not consistent with hepatic steatosis (which is usually defined by an attenuation level of <40 HU).² A nonspecific, round 3.3-cm hypodense lesion at the upper aspect of the spleen was noted.

A diagnostic test was performed.



Figure 1. Clinical Photographs.

Clinical photographs obtained at the time of admission to this hospital show acromegaly-like features on the face, including broadening of the nose, thickening of the lips, and skin tags (Panel A); extensive acanthosis nigricans involving the neck and axillae (Panel B); and skin breakdown, muscle atrophy, and low fat mass in the legs (Panel C).

Table 1. Laboratory Data.*

Variable	Reference Range, Adults, This Hospital†	On Admission, This Hospital
Hematocrit (%)	36.0–46.0	20.7
Hemoglobin (g/dl)	12.0–16.0	6.8
White-cell count (per μ l)	4500–11,000	1460
Differential count (per μ l)		
Neutrophils	1800–7700	949
Lymphocytes	1000–4800	394
Monocytes	200–1200	103
Eosinophils	0–900	14
Basophils	0–300	0
Platelet count (per μ l)	150,000–400,000	159,000
Mean corpuscular volume (fl)	80.0–100	91.4
Mean corpuscular hemoglobin (pg)	26.0–34.0	29.0
Mean corpuscular hemoglobin concentration (g/dl)	31.0–37.0	31.8
Red-cell distribution width (%)	11.5–14.5	15.5
Reticulocyte count (%)	0.7–2.5	1.5
Prothrombin time (sec)	11.5–14.5	13.3
Prothrombin-time international normalized ratio	0.9–1.1	1.0
Activated partial-thromboplastin time (sec)	22.0–36.0	27.9
Sodium (mmol/liter)	135–145	132
Potassium (mmol/liter)	3.4–5.0	3.7
Chloride (mmol/liter)	98–108	101
Carbon dioxide (mmol/liter)	23–32	23
Urea nitrogen (mg/dl)	8–25	24
Creatinine (mg/dl)	0.60–1.50	0.74
Glucose (mg/dl)	70–100	311
Calcium (mg/dl)	8.5–10.5	9.0
Albumin (g/dl)	3.3–5.0	3.3
Cholesterol (mg/dl)		
Total	<200	106
High-density lipoprotein	35–100	62
Low-density lipoprotein	50–129	21
Non-high-density lipoprotein	—	44
Triglycerides (mg/dl)	40–150	113
Glycated hemoglobin (%)	4.3–5.6	11.9
C peptide (ng/ml)	1.1–4.4	12.1
Glucagon (pg/ml)	<80	79
Proinsulin (pmol/liter)	3.6–22.0	8.8
Insulin-like growth factor 1 (ng/ml)	52–328	58
Thyrotropin (μ IU/ml)	0.40–5.00	1.18
Testosterone (ng/dl)	<50	<12

Table 1. (Continued.)

Variable	Reference Range, Adults, This Hospital†	On Admission, This Hospital
Sex hormone-binding globulin (nmol/liter)	21–139	105

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for cholesterol to millimoles per liter, multiply by 0.02586. To convert the values for triglycerides to millimoles per liter, multiply by 0.01129. To convert the values for insulin-like growth factor 1 to nanomoles per liter, divide by 7.7.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

DIFFERENTIAL DIAGNOSIS

Dr. Lindsay T. Fourman: I participated in the care of this patient, and I am aware of the final diagnosis. The differential diagnosis presented here reflects my initial assessment of the patient at the time of her transfer to our hospital. This 46-year-old woman had received a diagnosis of type 2 diabetes 5 years before the current presentation. In the time between the diagnosis and the current presentation, the doses of exogenous insulin that were needed to lower her blood glucose level progressively increased. This impaired response to glucose-lowering medications is consistent with insulin resistance.

Physiologic insulin secretion approximates 0.4 to 1 unit per kilogram of body weight per day.³ An exogenous insulin requirement greater than 2 or 3 units of insulin per kilogram per day is classified as severe insulin resistance.⁴ On transfer to this hospital, this patient had a strikingly high insulin requirement, which exceeded 50 units per kilogram per day.

INSULIN PSEUDORESISTANCE

In patients with suspected insulin resistance, I first consider potential mimics. Poor adherence to insulin therapy or excessive carbohydrate consumption may masquerade as reduced insulin sensitivity. In addition, when insulin is administered subcutaneously, scar tissue or soft-tissue edema at injection sites may result in decreased insulin delivery to peripheral tissues.⁵ Furthermore, the use of exogenous insulin that is expired or not stored properly (e.g., exposed to sunlight or not refrigerated) may reduce efficacy. This patient received high doses of insulin both intravenously and subcutaneously while in a monitored hospital

setting, which made insulin pseudoresistance an unlikely diagnosis. In addition, she had classic signs and symptoms of insulin resistance that occur due to cross-reactivity of high circulating levels of insulin with the insulin-like growth factor 1 (IGF-1) receptor, including acanthosis nigricans, skin tags, and acromegaly-like features (Fig. 1).⁴

PHYSIOLOGIC AND PATHOPHYSIOLOGICAL CONDITIONS

Puberty, pregnancy, and older age are risk factors associated with insulin resistance, but none of these applied to this patient. Insulin resistance can also occur in association with pathophysiological conditions, such as acute stress (e.g., trauma or sepsis), end-organ damage (e.g., cirrhosis or uremia), weight gain, obesity, or starvation.⁶ This patient had obesity when she received her initial diabetes diagnosis. However, obesity was an unlikely explanation for this patient's presentation because her insulin requirements continued to escalate despite profound weight loss and were highest when her BMI was in the underweight range.

MEDICATIONS

Specific medications may contribute to a reduced responsiveness to the glucose-lowering effect of insulin, including glucocorticoids, atypical antipsychotic agents, and antiretroviral therapy for human immunodeficiency virus infection.⁷ However, this patient did not receive any of these medications.

ENDOCRINE DISORDERS

Glucagonoma, Cushing's syndrome, and acromegaly are endocrine disorders involving hormonal hypersecretion that can lead to insulin resistance.⁶

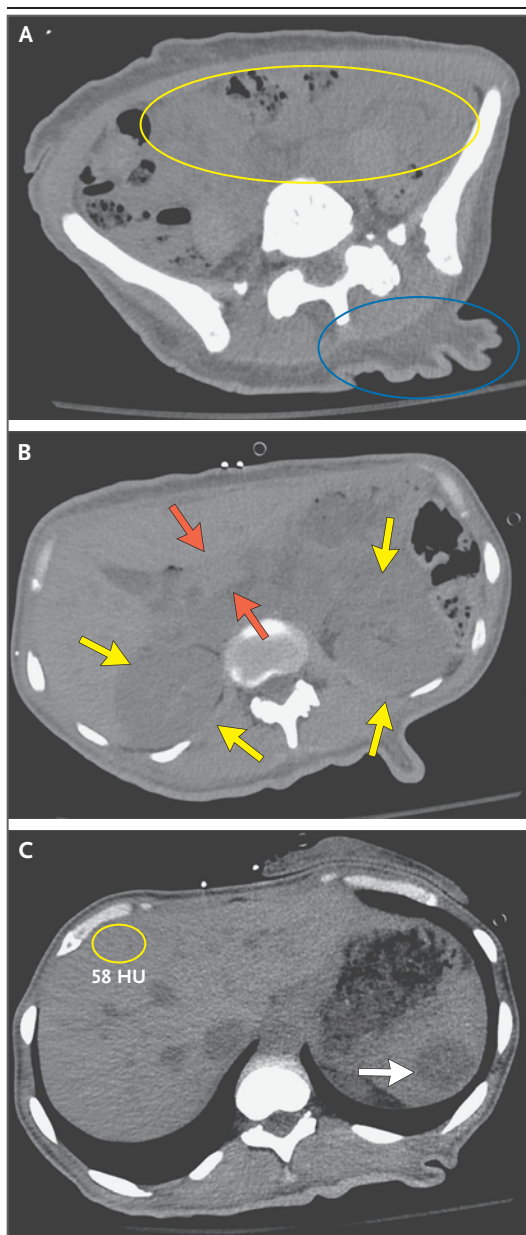


Figure 2. CT of the Abdomen and Pelvis.

CT was performed without the administration of contrast material. An axial image at the level of the mid-pelvis (Panel A) shows a near-complete absence of visceral fat between loops of nondilated bowel (yellow oval) and multiple skin folds with a near-complete absence of subcutaneous fat (blue oval). An axial image at the level of the mid-abdomen (Panel B) shows an absence of perinephric fat (yellow arrows) and pancreatic atrophy (red arrows). An axial image at the level of the upper abdomen (Panel C) shows a 3.3-cm hypodense lesion within the spleen (arrow) and a liver attenuation level of 58 Hounsfield units (HU) (oval).

Glucagonoma is a neuroendocrine tumor of the pancreatic islets that secretes glucagon. Hyperglycemia and weight loss are common findings in patients with glucagonoma. However, necrolytic migratory erythema, a characteristic rash present in more than half of patients with glucagonoma, was not seen in this case.⁸ In addition, the plasma level of glucagon was normal.

Cushing's syndrome is characterized by a constellation of signs and symptoms associated with prolonged exposure to elevated levels of circulating cortisol, which may occur in patients who receive glucocorticoids or have pituitary or adrenal tumors. However, the typical findings associated with Cushing's syndrome — such as weight gain with central adiposity, rounded facies, fatigue, proximal muscle weakness, easy bruising, and violaceous striae — were not present in this case.

Acromegaly, a condition resulting from excessive secretion of growth hormone produced in the anterior pituitary gland, is most often caused by a pituitary tumor. Although this patient presented with some facial features consistent with acromegaly (i.e., broadening of the nose and thickening of the lips), there was no acral enlargement, joint pain, headache, or sweating, all of which are characteristic of this condition. In addition, the serum level of IGF-1 was not elevated in this patient.

Menstrual irregularities may develop in patients with Cushing's syndrome or acromegaly. However, I suspected that this patient's amenorrhea was most likely related to her low BMI. Thyrotoxicosis and pheochromocytoma are other endocrine disorders associated with insulin resistance, although this patient's clinical presentation was not suggestive of these diagnoses.

LIPODYSTROPHIC SYNDROMES

Lipodystrophy refers to a heterogeneous group of rare disorders characterized by a selective lack of subcutaneous adipose tissue accompanied by insulin resistance.^{9,10} Lipodystrophic syndromes can be classified by the extent of deficient fat (generalized vs. partial) and the cause (genetic vs. acquired). In cases of partial lipodystrophy, areas of lipohypertrophy may also be present, involving sites such as the abdomen, neck, and chin. Insulin resistance develops in patients with lipodystrophy as a result of elevated circulating

free fatty acids and the accumulation of ectopic fat. Low fat stores also may lead to low circulating levels of the adipokine leptin, which further contributes to insulin resistance.⁹

When evaluating a patient with severe insulin resistance, it is important to differentiate lipodystrophy from cachexia due to poorly controlled diabetes.⁹ Insulin is an anabolic hormone, and thus impaired insulin signaling in patients with insulin resistance may lead to catabolism of both muscle and fat. By contrast, patients with lipodystrophy have reduced body fat with preservation of muscle mass. On examination, this patient had muscle atrophy and low body fat, findings consistent with cachexia. However, a lack of subcutaneous fat accompanied by a low circulating level of leptin can occur in patients with either of these conditions. Thus, lipodystrophy remained a diagnostic possibility in my evaluation of this patient.

PRIMARY INSULIN-SIGNALING DEFECTS

Primary insulin-signaling defects are rare genetic or acquired disorders in which insulin signaling is directly impaired upstream, at the level, or downstream of the insulin receptor (Fig. 3).

Primary insulin-signaling defects that occur upstream of the insulin receptor (prereceptor insulin-signaling defects) are caused by insulin antibodies that bind to circulating insulin.¹¹ Insulin antibodies impair the ability of insulin to interact with its receptor, which leads to postprandial hyperglycemia. Fasting hypoglycemia may also occur owing to the gradual dissociation of insulin from its antibodies. Acanthosis nigricans does not typically develop in patients with insulin antibodies, since bound insulin does not cross-react with the IGF-1 receptor. This patient had severe acanthosis nigricans without fasting hypoglycemia, which made prereceptor insulin resistance an unlikely diagnosis.

Primary insulin-signaling defects that occur at the level of the insulin receptor (receptor-level insulin-signaling defects) are caused by mutations in the receptor or by antibodies that target the receptor.^{4,10} Conditions that affect the insulin receptor may manifest at birth as autosomal recessive syndromes associated with growth and developmental disorders. Alternatively, heterozygous or homozygous mutations in the insulin

receptor may result in type A insulin resistance syndrome, which often becomes clinically apparent during or after adolescence. Type B insulin resistance syndrome is a rare disorder characterized by the presence of antibodies to the insulin receptor. This disorder most commonly occurs in middle-aged Black women who have other autoimmune features or lymphoproliferative disorders.¹²

Primary insulin-signaling defects that occur downstream of the insulin receptor (postreceptor insulin-signaling defects) rarely arise from mutations involving the insulin-signaling cascade.⁴ More commonly, postreceptor insulin resistance is caused by functional inhibition of intermediates within the insulin-signaling pathway (such as by posttranslational modification), which underlies the classic form of insulin resistance observed in patients with conditions such as obesity and lipodystrophy.^{13,14}

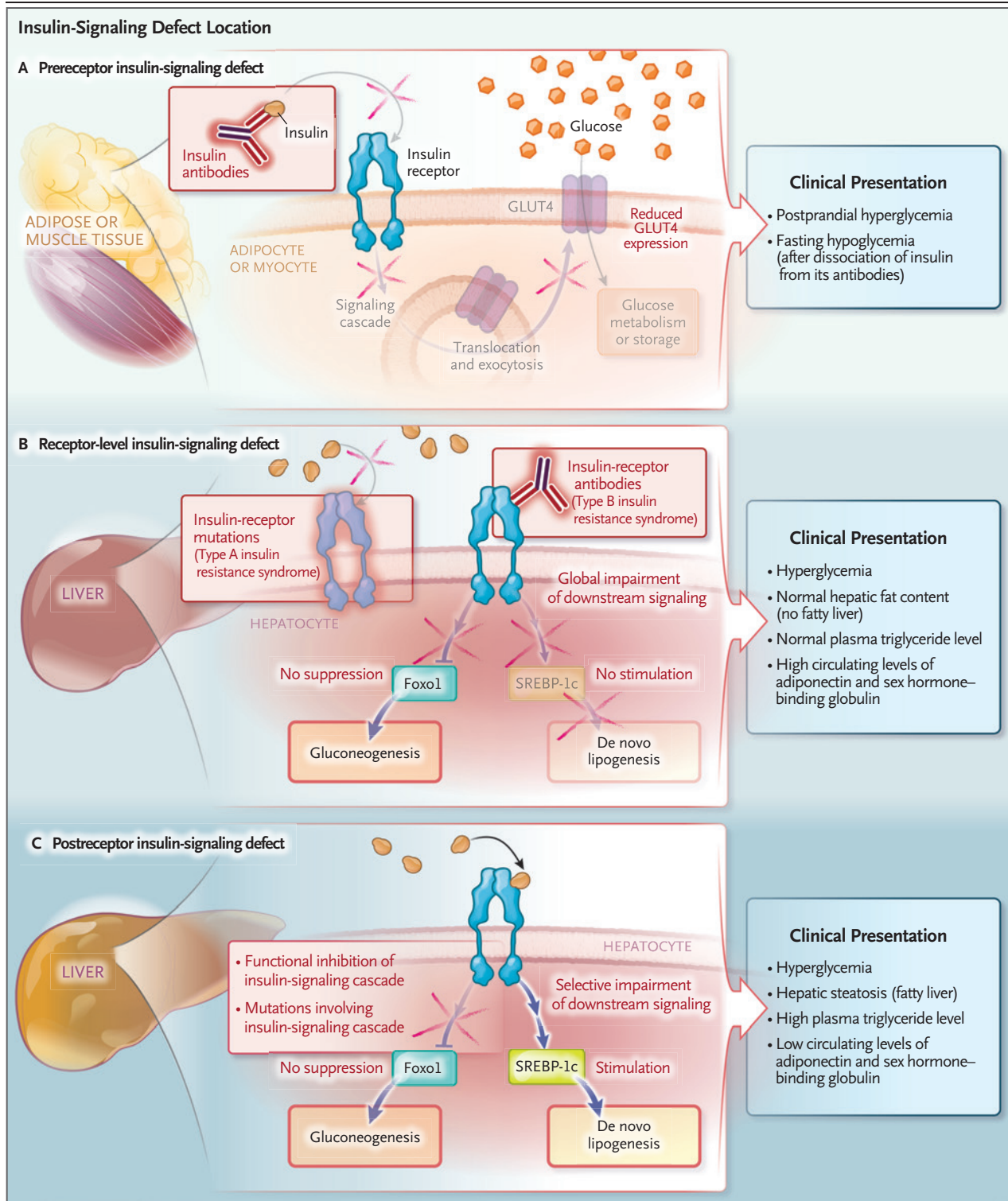
This patient's demographic characteristics and autoimmune profile were suggestive of a diagnosis of type B insulin resistance syndrome due to insulin-receptor antibodies. However, lipodystrophy remained a diagnostic consideration in this case.

DISTINGUISHING TYPE B INSULIN RESISTANCE SYNDROME FROM LIPODYSTROPHY

In the liver, the insulin-signaling cascade branches downstream of its receptor to form one pathway that suppresses gluconeogenesis and another pathway that activates *de novo* lipogenesis (Fig. 3). When insulin resistance is caused by a postreceptor insulin-signaling defect, as in the case of lipodystrophy, the suppression of gluconeogenesis is selectively impaired, whereas activation of *de novo* lipogenesis remains intact.¹³ The inability of insulin to suppress gluconeogenesis results in hyperglycemia. Meanwhile, upregulation of *de novo* lipogenesis by high circulating levels of insulin leads to hepatic steatosis. Accordingly, hepatic steatosis and hypertriglyceridemia are common features of postreceptor insulin resistance in patients with lipodystrophy.^{9,15} By contrast, receptor-level insulin resistance, as in the case of type B insulin resistance, results in global impairment of insulin signaling downstream of its receptor. Suppression of gluconeogenesis and activation of *de novo* lipogenesis both fail to occur in response to insulin, which leads to the

rare combination of insulin-resistant diabetes, normal hepatic fat content, and a normal plasma level of triglycerides.^{13,15} Receptor-level insulin resistance also results in high or high-normal

circulating levels of adiponectin and sex hormone-binding globulin, which are typically suppressed by partially intact insulin signaling in postreceptor insulin resistance.^{16,17}



This patient had normal hepatic fat content and a normal triglyceride level, as well as high-normal levels of adiponectin and sex hormone-binding globulin. These findings made a diagnosis of lipodystrophy unlikely and instead favored a diagnosis of type B insulin resistance syndrome due to insulin-receptor antibodies.

DR. LINDSAY T. FOURMAN'S DIAGNOSIS

Type B insulin resistance syndrome.

Figure 3 (facing page). Localization of Insulin Resistance at the Cellular Level.

Insulin resistance may occur upstream, at the level, or downstream of the insulin receptor. The differential biologic characteristics of each form of insulin resistance underlie its unique clinical presentation. Although defective insulin signaling occurs across multiple organ systems, the tissue or tissues most salient to distinguishing each form of insulin resistance from other causes are featured here. Prereceptor insulin resistance (Panel A) arises from antibodies that bind to circulating insulin. These antibodies transiently block the ability of insulin to interact with its receptor, leading to impaired glucose uptake by adipose tissue and muscle with consequent postprandial hyperglycemia. Subsequently, the gradual dissociation of insulin from these antibodies leads to a delayed effect of insulin on glucose uptake, which results in fasting hypoglycemia. Receptor-level insulin resistance (Panel B) arises from mutations or antibodies to the insulin receptor. This form of insulin resistance is characterized by a global impairment of insulin signaling downstream of its receptor, such that insulin fails to suppress gluconeogenesis or to stimulate de novo lipogenesis in the liver. Accordingly, receptor-level insulin-signaling defects lead to insulin-resistant diabetes in conjunction with uniquely normal hepatic fat content and a normal plasma triglyceride level. Receptor-level insulin resistance also results in high or high-normal circulating levels of adiponectin and sex hormone-binding globulin, which are typically suppressed by partially intact insulin signaling in postreceptor insulin resistance. Postreceptor insulin resistance (Panel C), which underlies conditions such as obesity and lipodystrophy, most commonly arises from the functional inhibition of intermediates within the insulin-signaling cascade (e.g., by posttranslational modification). More rarely, mutations involving the insulin-signaling cascade can result in this condition. Postreceptor insulin resistance is characterized by a selective impairment of insulin signaling. In the liver, insulin fails to suppress gluconeogenesis, yet stimulation of de novo lipogenesis by high circulating levels of insulin remains intact. The partial nature of postreceptor insulin resistance gives rise to a constellation of features characteristic of the metabolic syndrome, including hyperglycemia, hypertriglyceridemia, and hepatic steatosis.

DIAGNOSTIC TESTING

Dr. Stephen O'Rahilly: Diagnostic testing for insulin-receptor antibodies can be performed by means of immunoprecipitation followed by Western blot analysis. In this case, at a serum dilution of 1:5, there was a strong positive signal for the presence of insulin-receptor antibodies (Fig. 4). This finding, in combination with this patient's clinical presentation, is consistent with a diagnosis of type B insulin resistance syndrome, a condition that was first reported in 1976.¹⁸

Until recently, there were no commercially available assays for the measurement of insulin-receptor antibodies. However, in 2023, a new assay was developed that was based on a bridge assay detection format, which captures insulin-receptor antibodies between an insulin receptor containing a C-terminal peptide tag bound in the solid phase and a soluble insulin receptor with luciferase encoded at its C-terminus.¹⁹ The rarity of type B insulin resistance syndrome may present challenges for the widespread availability of such

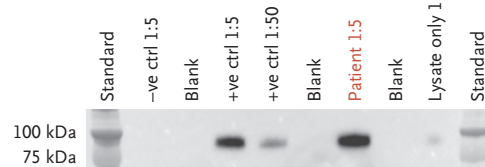


Figure 4. Western Blot Analysis.

Lysate from CL6 cells stably expressing the human insulin receptor was added to a sample of the patient's serum in addition to positive and negative controls. The Western blot image analysis confirmed the presence of insulin-receptor antibodies in the patient's serum. The insulin receptor β subunit was detected at 95kDa with the use of immunoprecipitation, shown as dark bands in the lanes for positive controls (+ve ctrl 1:5 and 1:50) and patient serum and as a faint band in the lysate-only lane. The standard lanes contained protein markers of a known size. The blank lanes contained diluent only. The positive control lanes contained serum from a patient who had had a diagnosis of type B insulin resistance syndrome; the serum was diluted at a ratio of 1:5 or 1:50 and incubated with recombinant human insulin receptors. The negative control (-ve ctrl) lane contained serum specimens from a patient who had not been found to have type B insulin resistance syndrome; the serum was diluted at a ratio of 1:5 and incubated with recombinant human insulin receptors. The lysate-only lane contained lysate from CL6 cells stably expressing the human insulin receptor.

an assay, but this assay represents a potentially substantial advance in facilitating the diagnosis of type B insulin resistance syndrome — a serious but treatable disorder.

LABORATORY DIAGNOSIS

Type B insulin resistance syndrome.

DISCUSSION OF MANAGEMENT

Dr. Rebecca J. Brown: The goal of treatment for type B insulin resistance syndrome is the complete remission of diabetes resulting from the elimination of the insulin-receptor antibody. Spontaneous remission can occur in some patients, but the time to remission may take many years. When left untreated, type B insulin resistance syndrome is associated with a mortality rate close to 50%.¹² Owing to the rarity of this condition, data from randomized, controlled trials of immunosuppressive or antibody-depletion therapies for the treatment of type B insulin resistance syndrome are lacking. A variety of treatment approaches have been tested in small numbers of patients, including plasmapheresis and plasma exchange,²⁰ intravenous immune globulin,²¹ and immunosuppressive agents (e.g., rituximab, glucocorticoids, azathioprine, cyclophosphamide, and mycophenolate mofetil).^{12,22-25} In 2010, Malek and colleagues reported on the successful treatment of type B insulin resistance syndrome in seven patients who received a combination of the following immunosuppressive agents: rituximab (an anti-CD20 monoclonal antibody that was used to target antibody-producing B lymphocytes), high-dose glucocorticoid pulse therapy (used to reduce the activity of antibody-containing plasma cells), and cyclophosphamide or cyclosporine (nonspecific B-cell and T-cell-suppressing drugs). The patients were switched to maintenance therapy with azathioprine once remission occurred.²⁶ In a follow-up report involving 22 patients, this combination regimen led to remission (defined as amelioration of the hyperglycemia with discontinuation of insulin treatment or normalization of hyperandrogenemia [or both]) in 19 of the patients after a mean duration of 5 months.²⁷ Toxic effects related to the use of this combination regimen of immunosuppressive agents vary depending on the individual agents used and include leukopenia,

increased risk of infection, gonadal dysfunction, and temporary exacerbation of insulin resistance from glucocorticoids.

The induction of remission of diabetes in patients with type B insulin resistance syndrome usually takes months; therefore, effective management of diabetes while awaiting remission is a critical component of care. Owing to the extreme nature of type B insulin resistance syndrome, the standard inpatient or outpatient targets for blood glucose control are not feasible. Instead, the goals of diabetes management are to reverse the catabolic state observed in most patients and to allow patients to be safely treated with immunosuppressive therapy, while reducing the risk of both diabetic ketoacidosis and hypoglycemia.

The median dose of insulin used in the case series involving 22 patients with type B insulin resistance syndrome was 1775 units per day; doses as high as 18,000 units per day were reported.²⁷ In such cases, concentrated insulin products, including human regular U-500 insulin, can be used to reduce the injection burden. Antibodies block the insulin receptor in patients with type B insulin resistance syndrome, so insulin is not cleared normally through receptor-mediated degradation.²⁷ As a result, both endogenous and exogenous insulin have prolonged circulating half-lives, and it is not possible to replicate normal insulin physiology with the use of a basal-bolus insulin regimen. To reduce the risk of nocturnal hypoglycemia and fasting hypoglycemia due to a prolonged insulin half-life, most patients should receive the highest doses of insulin in the morning, followed by progressively lower doses over the course of the day. Hypoglycemia develops in some patients even in the absence of exogenous insulin due to the agonistic effects of low-titer antibodies at the insulin receptor. In these cases, the management of the hypoglycemia usually involves glucocorticoids or other treatments for ongoing immunosuppression,²⁸ and sometimes treatment is supplemented with slow-release carbohydrates (e.g., uncooked cornstarch).²⁹

Dr. Shank: Additional testing was performed in this patient on the basis of the identification of insulin-receptor antibodies and the suspicion of an underlying autoimmune disease. Tests for antibodies to ribonucleoprotein were highly positive, and these findings were thought to be consistent with a diagnosis of systemic lupus erythematosus or mixed connective-tissue disease. A

biopsy of the splenic lesion was performed, which led to a diagnosis of lymphangioma.

Immunomodulatory therapy with rituximab, cyclophosphamide, pulse glucocorticoids, and hydroxychloroquine was initiated. Drug-induced liver injury developed and was attributed to cyclophosphamide, which was switched to intravenous immunoglobulin and azathioprine.

Unfortunately, after initial treatment at this hospital, the patient had recurrent hospitalizations for complications related to immunosuppression, including esophageal candidiasis, *Clostridium difficile* colitis, and reactivation of pulmonary blastomycosis. These infections led to multiple interruptions in her immunomodulatory therapy. Six months after

the initial transfer to this hospital, she is still receiving high-dose human regular U-500 insulin, but her weight has increased by 18 kg, and the glycated hemoglobin level has decreased by 2.4%. The patient continues to undergo treatment with immunomodulatory agents and is being monitored for treatment-related toxic effects, with the goal of complete remission of type B insulin resistance syndrome.

FINAL DIAGNOSIS

Type B insulin resistance syndrome.

This case was presented at Medicine Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

REFERENCES

1. Macauley M, Percival K, Thelwall PE, Hollingsworth KG, Taylor R. Altered volume, morphology and composition of the pancreas in type 2 diabetes. *PLoS One* 2015;10(5):e0126825.
2. Lawrence DA, Oliva IB, Israel GM. Detection of hepatic steatosis on contrast-enhanced CT images: diagnostic accuracy of identification of areas of presumed focal fatty sparing. *AJR Am J Roentgenol* 2012;199:44-7.
3. Janež A, Guja C, Mitrakou A, et al. Insulin therapy in adults with type 1 diabetes mellitus: a narrative review. *Diabetes Ther* 2020;11:387-409.
4. Semple RK, Savage DB, Cochran EK, Gorden P, O'Rahilly S. Genetic syndromes of severe insulin resistance. *Endocr Rev* 2011;32:498-514.
5. Ariza-Andraca CR, Altamirano-Bustamante E, Frati-Munari AC, Altamirano-Bustamante P, Graef-Sánchez A. Delayed insulin absorption due to subcutaneous edema. *Arch Invest Med (Mex)* 1991;22:229-33.
6. Tritos NA, Mantzoros CS. Clinical review 97: syndromes of severe insulin resistance. *J Clin Endocrinol Metab* 1998;83:3025-30.
7. Fève B, Scheen AJ. When therapeutic drugs lead to diabetes. *Diabetologia* 2022;65:751-62.
8. Wermers RA, Fatourehchi V, Wynne AG, Kvols LK, Lloyd RV. The glucagonoma syndrome: clinical and pathologic features in 21 patients. *Medicine (Baltimore)* 1996;75:53-63.
9. Fourman LT, Grinspoon SK. Approach to the patient with lipodystrophy. *J Clin Endocrinol Metab* 2022;107:1714-26.
10. Angelidi AM, Filippaios A, Mantzoros CS. Severe insulin resistance syndromes. *J Clin Invest* 2021;131(4):e142245.
11. Greenfield JR, Tuthill A, Soos MA, et al. Severe insulin resistance due to anti-insulin antibodies: response to plasma exchange and immunosuppressive therapy. *Diabet Med* 2009;26:79-82.
12. Arioglu E, Andewelt A, Diabo C, Bell M, Taylor SI, Gorden P. Clinical course of the syndrome of autoantibodies to the insulin receptor (type B insulin resistance): a 28-year perspective. *Medicine (Baltimore)* 2002;81:87-100.
13. Semple RK, Sleight A, Murgatroyd PR, et al. Postreceptor insulin resistance contributes to human dyslipidemia and hepatic steatosis. *J Clin Invest* 2009;119:315-22.
14. Samuel VT, Shulman GI. Mechanisms for insulin resistance: common threads and missing links. *Cell* 2012;148:852-71.
15. Mann JP, Savage DB. What lipodystrophies teach us about the metabolic syndrome. *J Clin Invest* 2019;129:4009-21.
16. Semple RK, Soos MA, Luan J, et al. Elevated plasma adiponectin in humans with genetically defective insulin receptors. *J Clin Endocrinol Metab* 2006;91:3219-23.
17. Semple RK, Cochran EK, Soos MA, et al. Plasma adiponectin as a marker of insulin receptor dysfunction: clinical utility in severe insulin resistance. *Diabetes Care* 2008;31:977-9.
18. Flier JS, Kahn CR, Roth J, Bar RS. Antibodies that impair insulin receptor binding in an unusual diabetic syndrome with severe insulin resistance. *Science* 1975;190:63-5.
19. Minich WB, Abel BS, Schwiebert C, et al. A novel in vitro assay correlates insulin receptor autoantibodies with fasting insulin in type B insulin resistance. *J Clin Endocrinol Metab* 2023;108:2324-9.
20. Page KA, DeJardin S, Kahn CR, Kulkarni RN, Herold KC, Inzucchi SE. A patient with type B insulin resistance syndrome, responsive to immune therapy. *Nat Clin Pract Endocrinol Metab* 2007;3:835-40.
21. Zhang S, Wang G, Wang J. Type B insulin resistance syndrome induced by systemic lupus erythematosus and successfully treated with intravenous immunoglobulin: case report and systematic review. *Clin Rheumatol* 2013;32:181-8.
22. Kawashiri S-Y, Kawakami A, Fujikawa K, et al. Type B insulin resistance complicated with systemic lupus erythematosus. *Intern Med* 2010;49:487-90.
23. Coll AP, Morganstein D, Jayne D, Soos MA, O'Rahilly S, Burke J. Successful treatment of Type B insulin resistance in a patient with otherwise quiescent systemic lupus erythematosus. *Diabet Med* 2005;22:814-5.
24. Coll AP, Thomas S, Mufti GJ. Rituximab therapy for the type B syndrome of severe insulin resistance. *N Engl J Med* 2004;350:310-1.
25. Yang H, Zhao J, Li Y, Lv F, Zhang S, Li Y. Successful treatment of type B insulin resistance with mixed connective tissue disease by pulse glucocorticoids and cyclophosphamide. *J Diabetes Investig* 2017;8:626-8.
26. Malek R, Chong AY, Lupsa BC, et al. Treatment of type B insulin resistance: a novel approach to reduce insulin receptor autoantibodies. *J Clin Endocrinol Metab* 2010;95:3641-7.
27. Klubo-Gwiedzinska J, Lange M, Cochran E, et al. Combined immunosuppressive therapy induces remission in patients with severe type B insulin resistance: a prospective cohort study. *Diabetes Care* 2018;41:2353-60.
28. Brown N, Elston MS. Systematic review — type B insulin resistance with isolated hypoglycemia and suppressed insulin. *J Clin Endocrinol Metab* 2023 August 8 (Epub ahead of print).
29. Censi S, Mian C, Betterle C. Insulin autoimmune syndrome: from diagnosis to clinical management. *Ann Transl Med* 2018;6:335.

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